Field2VC: Automating the Generation of ASPICE-Compliant Verification Criteria from Field Failures

Anonymous Author(s)
Submission under double-blind review
Affiliation withheld for anonymity

***Abstract*—In Automotive SPICE, every SYS.2 system requirement must be verifiable through explicit verification criteria (VCs). Yet writing VCs that anticipate real-world field failures is slow and systematically incomplete: engineers rarely foresee failure modes they have not previously encountered. We present Field2VC, which derives VCs from two complementary failure records—developer-reported defects on GitHub, which expose code-level boundary conditions absent from specifications, and consumer-reported complaints in the NHTSA ODI database, which expose vehicle-level failures engineers do not anticipate. Field2VC extracts each report into a structured failure representation, retrieves the failures most relevant to a given requirement, and prompts a large language model to convert this evidence into specific, testable VCs. We contribute (i) an industrial benchmark of 350 de-identified SYS.2 requirements with expert-written reference VCs, 307 of which entered evaluation, and (ii) an empirical study in which four automotive domain experts assessed 489 generated VCs. Grounding generation in field failures raises expert acceptance from a 45.3 % no-context baseline to 65.5 %–67.2 % across the two sources (odds ratio 2.3–2.5×, Holm-corrected $p < 0.001$; Cohen's $h = 0.41$–0.45, a small-to-medium effect). The gain is consistent across body-control, communication-stack, and diagnostics requirement categories.**

***Index Terms*—verification criteria, automotive requirements, ASPICE SYS.2, issue mining, LLM, failure knowledge, ISO 26262, NHTSA**

I. Introduction

In automotive software engineering, the cost of repairing a defect grows exponentially across the development lifecycle. Boehm and Basili's empirical survey finds that fixing a defect after delivery can cost up to 100 times more than fixing it during the requirements phase [1] (Fig. 1). In addition, ASPICE SYS.2 strictly requires every system requirement to be *verifiable*: a *Verification Criterion* (VC) must coexist with the requirement text [2].

Writing thorough VCs, however, fails in a specific and repeatable way. A requirement specifies intended behaviour under nominal conditions; a complete VC must additionally probe boundary conditions and fault modes. Reviewers reason forward from the requirement text, so the criteria they write can only confirm behaviours the specification already anticipates, while field failures, almost by definition, occur where it did not. The blind spot is structural rather than a matter of effort: a VC that omits an unforeseen edge case passes internal review precisely because no reviewer holds the evidence that would expose it (Fig. 1).

Closing this gap requires failure evidence from outside the project. Defect trackers are a natural candidate, and recent work such as CrUISE-AC [3] mines them to enrich agile acceptance criteria. For ASPICE SYS.2, however, the evidence must match the abstraction level at which the requirement is verified: the *system* level, not the software unit. Developer-reported defects describe faults as engineers observe them: isolated to a module, reproducible, often with a root cause attached. Failures that emerge from interactions across subsystems, environments, and months of customer usage rarely surface in any repository, because no developer observes the deployed fleet. Regulator-collected complaints capture precisely this stratum. NHTSA's ODI database records faults as the vehicle exhibits them, at fleet scale, under operating conditions no test plan enumerates [4], [5]. The two corpora are therefore not interchangeable: they sample the failure space at different abstraction levels. Our evaluation confirms this split empirically—complaint-grounded VCs are accepted more often for body and chassis requirements, while defect-grounded VCs lead for communication and diagnostics modules (§V-B). We use NHTSA as a public proxy for the warranty and field-return data an OEM holds internally; the pipeline itself is agnostic to where the vehicle-level channel comes from. Manual inspection of 200 randomly sampled requirement–evidence pairs confirms the reliability of this evidence-mapping stage, yielding a precision of **73.5 %** (§IV).

This motivates our central question: given a system requirement, can generation grounded in cross-level field-failure evidence produce verification criteria that practising engineers would accept into a SYS.2 baseline? We treat expert acceptance as the measurable proxy for usefulness, since no oracle for "complete" VC coverage exists.

This paper presents **Field2VC**, an automated VC generation mechanism grounded in real failure evidence. Field2VC extracts defect context from two evidence channels: GitHub, which captures code-level software failures, and NHTSA, which captures vehicle-level failures. It filters domain-specific noise out of the raw records and uses the retained context as a semantic constraint, so that a large language model (LLM) generates verifiable VCs that cover historical failure scenarios.

a) Contributions.:

- **An automated, evidence-grounded pipeline** that combines developer-reported defects (GitHub) and regulator-

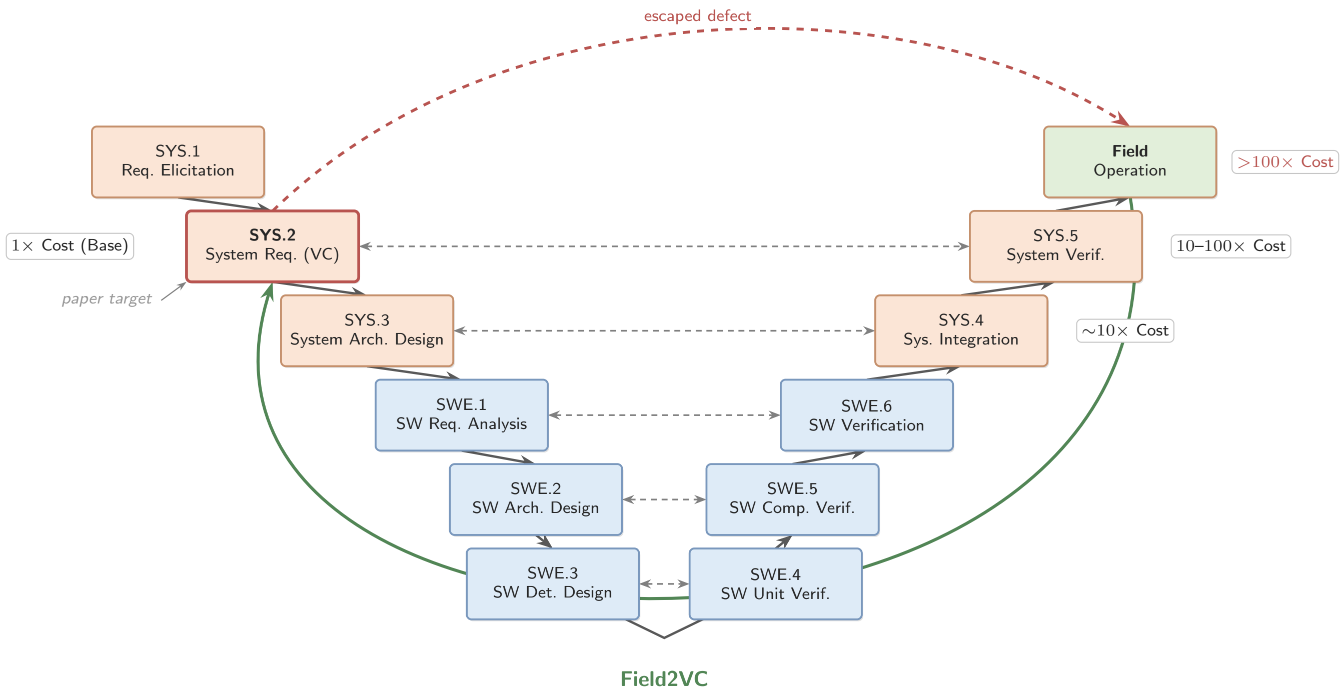

Fig. 1. ASPICE V-model cost escalation. A missing VC that passes SYS.2 review surfaces only at system-level testing or as a field complaint, where re-verification cost is 10–100× or more [1]. Field2VC shifts detection left by grounding VC generation in field-failure evidence *at* SYS.2 (highlighted by the *paper target* annotation); the green arc shows the evidence feedback loop from field operation back to SYS.2. SWE-layer processes (SWE.1–SWE.6) are outside Field2VC's scope: they receive the generated SYS.2 VCs as upstream constraints and are not modified by the pipeline.

collected complaints (NHTSA) as complementary evidence channels at distinct abstraction levels, with an abstention mechanism that declines to generate when retrieved evidence is insufficient (§III).

- **A four-expert evaluation.** Four domain experts (including one author) evaluated 489 generated VCs spanning 307 unique industrial requirements, showing that dual-channel evidence raises VC acceptance significantly above a no-context baseline (§V).
- **A quantitative characterisation of the two evidence channels**, showing a systematic division of labour: the complaint channel performs better on body and chassis systems, the defect channel on communication and diagnostics modules (§V-B).
- **An industrial benchmark dataset.** We open-source a de-identified dataset of 307 real automotive system requirements (every item that entered expert evaluation) with their expert-annotated golden VCs, together with all generated VCs, per-rater labels, and analysis scripts, to support replication (Data Availability).

Our results extend work on issue-tracker-driven requirements enrichment [3] in two directions that prior work does not cover: the evidence is drawn from two channels at *distinct abstraction levels*, and the target artefact is an ASPICE-mandated verification criterion for a safety-relevant ECU rather than an agile acceptance criterion.

b) Data Availability.: To support double-blind review with a complete chain of evidence, the de-identified replication package is available *now* for review at https://anonymous.4open.science/r/apsec2026-artifact-C562/, containing: (1) the de-identified SYRS corpus covering all 307 evaluated items (drawn from a 350-item benchmark), with their expert-annotated golden VCs; (2) all 489 generated VCs with per-VC rater labels and majority-vote accept/reject labels, plus the adjudication record of the coordinating expert; (3) the evaluation guidelines (Framework B); (4) pipeline scripts for Stages 2–5 and all prompt templates; and (5) statistical analysis scripts. Proprietary signal names are masked by stable placeholders; the redaction script itself is part of the package.

II. Related Work

A. Automated Requirements Completion within Specifications

Requirements completeness research falls into two streams. *Refinement* approaches improve individual requirements through semantic enrichment without adding new external data, using techniques such as NL-level templating and domain-knowledge-driven pattern systems, e.g., the EARS controlled requirements syntax [6]. *Derivation* approaches generate new requirements by exploiting relationships within or between existing requirement items.

LLM-based derivation has advanced rapidly [7]. ReqCompleter [8] fuses use-case models, E-R diagrams, and CRUD matrices into business-logic constraints that drive an iterative LLM completion process, achieving 20–88 % completeness improvement on seven real-world cases. F2SRD [9] retrieves the applicable OWASP ASVS verification requirements for each functional specification and guides GPT-4 to derive security requirements grounded in them. Both rely on *same-document* business logic as the completion signal; our work is complementary: we exploit *external* failure repositories to surface conditions that no single specification can reveal.

Closest in spirit is TVR [10], which applies retrieval-augmented generation (RAG) to validate and recover missing traceability links between stakeholder and system requirements in industrial automotive projects. TVR targets *traceability* (linking existing requirements) rather than *VC completion* (generating new complementary criteria); its retrieval corpus is the specification itself, whereas ours is external failure evidence. These two directions show that RAG-based requirements engineering is being applied across multiple automotive sub-tasks [11], [12].

B. Enriching Requirements from External Repositories

Mining bug reports and issue-tracking systems for requirements engineering has a long history [13], [14]. Do et al. [15], [16] extract relevant requirements from similar software products using doc2vec and BIRCH clustering. Almonte et al. [17] use LLM-based prompting to indirectly derive quality-attribute non-functional requirements from functional specifications. In a similar vein, Koscinski et al. [18] use RelGAN to transform functional information into security requirements. These approaches mine requirements-like data from whatever related corpora are available; we instead mine two repositories from *different domains*—a defect-tracking database and a regulator-operated consumer complaint database—and show that the choice of knowledge source strongly predicts VC quality.

CrUISE-AC [3] is the leading crowd-data-driven system over external repositories. It mines public issue trackers (seven e-commerce systems, >165 k issue records, plus two content-management systems in a second domain) for issues matching Gherkin-format user-story acceptance criteria, achieving an 82 % approval rate in the e-commerce domain under a five-expert majority-vote evaluation with Gwet's $\mathrm{AC1} = 0.74$. Wang et al. [19] likewise generate acceptance criteria from multi-modal requirements data with LLMs, but target user-facing acceptance criteria rather than safety-critical system requirements.

Field2VC differs from CrUISE-AC in three respects. First, our domain is the automotive system-requirements level (ASPICE SYS.2) for safety-relevant ECU components. Second, we evaluate against a three-criterion Framework B standard (Completeness ≥ 3, Novelty = Y, Usability = Y) rather than a binary acceptance judgement, a design that makes direct percentage comparisons misleading. Third, our ablation study contrasts LLM binary classification and embedding-based matching as grounding mechanisms for VC generation; it primarily exposes the effect of the filtering-enabled abstention decision on the results, rather than the matching method alone (§V-C).

C. LLM-based Requirements Engineering

Recent work uses LLMs for diverse requirements tasks [20]–[23]: MARE [24] proposes a multi-agent collaboration framework for requirements elicitation; LATO [25] applies layerwise LLM reasoning to generate UML activity diagrams from requirements. Neither targets VC completion, and neither evaluates the effect of grounding in external failure evidence, the key contribution of Field2VC. In the automotive-safety domain specifically, LLMs have been applied to code generation and verification [26]–[28], functional-safety and security design [29], and RAG-based derivation of safety requirements from standards [30]; these operate on code or upstream safety requirements rather than completing verification criteria for existing SYS.2 specifications.

III. Field2VC: Two-Channel VC Completion Pipeline

Field2VC operates as a five-stage pipeline that transforms raw failure records from two complementary external knowledge channels into candidate missing Verification Criteria (VCs) for automotive ECU SYRS items. Given a set of SYRS items and a target channel, the pipeline produces VCs grounded in empirical field-failure evidence, each traceable to one or more source records (Fig. 2). Throughout this section, we illustrate the pipeline using a real-world UDS Security Access requirement (SYRS.1018_B_B) and its corresponding field failure from GitHub (`python-udsoncan#51`). Table I presents two representative accepted VCs from the main evaluation—one per channel—selected to illustrate the complementary discovery roles of the two evidence sources: Channel A (GitHub) surfaces protocol-transport boundary conditions absent from specification, whereas Channel B (NHTSA) surfaces real-world electromechanical interaction scenarios that the golden VCs were not designed to anticipate.

A. Stage 1: Fetch

We define two channels providing complementary failure evidence:

Channel A: GitHub Issues. We index issues from four open-source repositories covering ECU-adjacent software stacks: `pylessard/python-udsoncan` (UDS diagnostics), `python-can-isotp` (ISO-TP transport), `hardbyte/python-can` (CAN bus), and `ecubus/EcuBus-Pro` (ECU test toolchain). We retrieved **696 issues** via the GitHub REST API,[1] prioritising closed issues labelled {bug, defect, regression, safety} and supplementing them with unlabelled closed issues for coverage. For the running example, this stage fetches `python-udsoncan#51`: *"SecurityAccess does not handle already-unlocked server."*

Channel B: NHTSA Complaints. We query the NHTSA ODI consumer-complaint database [31] using component keywords matching automotive ECU subsystems. We retrieved **364 complaint records**, each containing a free-text consumer

[1] https://docs.github.com/en/rest/issues

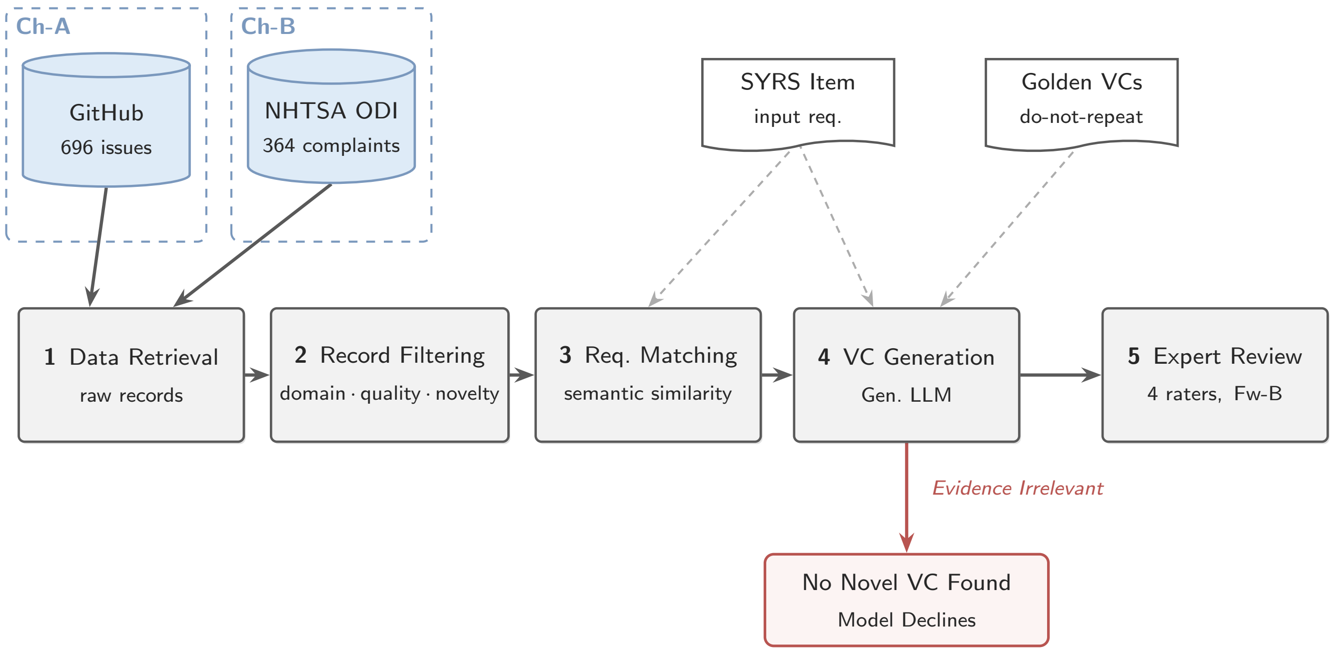

Fig. 2. Field2VC five-stage pipeline. Failure records from GitHub (Ch-A) and NHTSA (Ch-B) are filtered into ⟨Trigger | Protocol Behavior | Fault Mode⟩ triples (Stage 2, Qwen-Plus), embedding-matched to each SYRS item via cosine similarity (Stage 3, `text-embedding-v3`), and used to prompt VC generation (Stage 4, Qwen-Max). The generator is never forced to produce a VC: it may decline by emitting the fixed marker `NO_NOVEL_VC_FOUND` when the evidence reveals no genuinely missing condition, the pipeline's only relevance gate. Four domain experts review the surviving VCs under Framework B (Stage 5).

narrative plus structured vehicle and component metadata. Unlike curated recall notices, complaint narratives are raw, colloquial, and noisy field reports, which is why the Stage-2 filter matters most for this channel.

B. Stage 2: Filter

Raw failure text varies in structure, length, and vocabulary. We use an LLM (Qwen-Plus) to filter each record into a structured triple:

$$\langle \textit{Trigger} \mid \textit{Protocol Behavior} \mid \textit{Fault Mode} \rangle$$

Trigger is the activating condition (e.g., a specific message sequence or timing event), *Protocol Behavior* is the expected ECU action per the relevant standard, and *Fault Mode* is the observed deviation. Filtering reduces vocabulary mismatch and suppresses implementation-layer details irrelevant to SYS.2 requirements.

The prompt's core instruction is *"Extract the technical essence from this [issue/complaint] report"*, with a mandatory escape route: if the record is not about automotive system behaviour (e.g., pure Python packaging, documentation), the model must return all three fields as `null`.[2] Records whose three fields return `null` are dropped here; this is where off-topic noise (packaging problems, documentation requests, pure consumer venting) leaves the pipeline. For the running example, issue #51 reduces to: ⟨`0x27` requestSeed while already unlocked | client detects all-zero seed and skips sendKey | key derived from zero seed → NRC `0x24` rejection⟩.

[2] All Stage-2 and Stage-4 prompts are released verbatim in the replication package (`prompts/`).

C. Stage 3: Match

For each SYRS item s and each filtered triple t_i in the channel corpus, we compute cosine similarity [32]–[34] using the `text-embedding-v3` model (Alibaba DashScope; snapshot April–May 2026). We retrieve the top-$K = 5$ triples ranked by similarity, forming the *failure context* $\mathcal{F}(s) = \{t_1, \ldots, t_5\}$. We deliberately impose *no* hard similarity cut-off at this stage: every item proceeds with its top-5 context, and the relevance decision is delegated to the generation stage, where the model abstains (`NO_NOVEL_VC_FOUND`) if the retrieved evidence does not reveal a genuinely missing condition (§III-D). This mirrors, in post-retrieval form, the pre-retrieval relevance judgement that CrUISE-AC makes with an LLM binary classifier, a design choice our ablation study examines directly (RQ3, §V-C). For SYRS.1018_B_B, issue #51 ranks second at cosine 0.69 and enters the top-5 failure context.

Embedding-based matching provides two advantages over LLM binary classification (used in CrUISE-AC [3]): (1) it scales to thousands of (SYRS, issue) pairs without incurring per-pair LLM calls, and (2) it produces a continuous, auditable relevance score that is released with every match in our artefact.

TABLE I

WORKED TRACES: TWO REPRESENTATIVE ACCEPTED VCS FROM THE MAIN EVALUATION (ONE PER CHANNEL), TRACING THE FULL FIVE-STAGE CHAIN. FAILURE TEXT QUOTED VERBATIM FROM PUBLIC SOURCES; SYRS AND GOLDEN VCS PARAPHRASED FROM THE DE-IDENTIFIED CORPUS. CHANNEL A SURFACES A TRANSPORT-LAYER RECOVERY BOUNDARY ABSENT FROM SPECIFICATION; CHANNEL B A SYSTEM-LEVEL INTERACTION SCENARIO BEYOND THE NOMINAL-PATH GOLDEN VC.

Channel A: GitHub, SYRS.0719_A (UDS ECU Reset via DoIP). *SYRS:* implement UDS ECU Reset service (`0x11`, sub-function `hardReset 0x01`) and perform a software reset upon a valid request. The golden VC verifies *timing only*: positive response within 50 ms; software reset completion within 200 ms.

Matched failure: `udsoncan#159` (sim. 0.74): "Connection error after Reset—`[Errno 104] Connection reset by peer` / `[Errno 32] Broken pipe`" when reconnecting over DoIP after ECU reset.

Triple: ⟨ECU reset (`0x11 0x01`) issued over DoIP | ECU resets and DoIP transport recovers for subsequent communication | TCP connection terminates with RST or broken pipe; reconnection impossible⟩.

Generated VC: verify that DoIP transport-layer connectivity is re-established within protocol limits after a valid reset response is received—a post-reset reachability path absent from both the SYRS and its golden VC. **Accepted** (E2: C=5; E3: C=3; N = Y, U = Y; unanimous).

Channel B: NHTSA, SYRS.0177 (low-beam robustness under AC flicker). *SYRS:* do *not* switch off low-beam headlights when exposed to synthetic AC-flicker light sources if ambient brightness remains <1000 lux for more than 4 s. The golden VC tests the *nominal AC-flicker path only*.

Matched failure: ODI #11557628 (sim. 0.60; 2022 Toyota Camry): "low beam headlights flashing when high beams are on … also flash on ignition."

Triple: ⟨high-beam activation or ignition key-on | low-beam LEDs remain steadily illuminated | bilateral intermittent flashing (right front more severe); state-stuck fault⟩.

Generated VC: fault-injection in an AC-flicker environment ($<$1000 lux sustained $>$4 s): activate high beams and separately cycle the ignition, verifying that low beams neither flash nor extinguish in either case—electromechanical interaction scenarios absent from the golden VC. **Accepted** (E3: C=3; E4: C=3; N = Y, U = Y; unanimous).

D. Stage 4: Generate

We prompt Qwen-Max with three inputs: the SYRS item, its existing golden VCs (as *do-not-repeat* context that operationalises "missing"), and the failure context $\mathcal{F}(s)$. The prompt instructs the model to generate only VCs that (a) genuinely test the requirement (not the protocol library the failure was reported against), (b) are not redundant with the existing golden VCs, and (c) are inspired by a real failure pattern in $\mathcal{F}(s)$ and testable at ECU system integration level. The prompt's core instruction is *"generate MISSING verification conditions … 'Missing' means test scenarios NOT already covered by the existing golden VCs"*, and it closes with an explicit permission to decline: *"If no genuinely useful missing VC can be identified from these issues, output:* `NO_NOVEL_VC_FOUND`*."* We refer to this fixed marker as the *sentinel*. Declining is an expected outcome, not an error: the model is never forced to produce a VC, so when the retrieved evidence is irrelevant the pipeline abstains instead of hallucinating a plausible-sounding criterion [35]. In the running example, the model turns issue #51's triple into a negative test of the already-unlocked path ($27 L1 unlocked, send `$31 01 10 08`, expect no NRC $33)—a scenario neither the SYRS nor its golden VC exercises.

E. Stage 5: Review

Each generated VC is independently scored by at least two of the three external raters (E2, E3, E4) on three dimensions under **Framework B**:

- **C (Completeness)**: Score 1–5. Does the SYRS item omit this VC? ($C \geq 3$ is the acceptance threshold.)
- **N (Novelty)**: Y/N. Is this VC absent from the existing SYRS item and its associated VCs?
- **U (Usefulness)**: Y/N. Is this VC actionable for downstream test design and software qualification?

A VC is accepted if and only if $C \geq 3$ AND $N = \mathrm{Y}$ AND $U = \mathrm{Y}$. The label is the majority vote among the raters who scored each VC; E1 (adjudicator) resolves disagreements on the N and U dimensions. The running-example VC was accepted (C=4, N=Y, U=Y) after E1 adjudicated a novelty split between E2 and E3.

F. Ablation Variants

Two ablation conditions isolate the contribution of filtering and embedding-based matching:

Abl-NF (No Filter): Skip Stage 2. Embed raw issue title + body (first 500 characters) directly and match to SYRS via cosine similarity. This removes the structured triple extraction, testing whether raw-text embedding suffices.

Abl-LM (LLM-Match): Replace Stage 3 embedding similarity with an LLM binary relevance classifier (Qwen-Plus), replicating the CrUISE-AC matching approach [3]. This tests LLM judgement against embedding similarity as a pairing mechanism.

IV. EXPERIMENT DESIGN

A. Research Questions

- **RQ1 (Effectiveness):** Does failure-grounded prompting generate VCs with significantly higher expert acceptance than a no-context baseline?
- **RQ2 (Channel–Domain Fit):** The two channels are mapped to disjoint requirement subdomains by construction. Within each subdomain (body-control/chassis, communication-stack, diagnostics), how effective is the assigned channel relative to the no-context baseline, and how do the two channels divide labour across subdomains?
- **RQ3 (Ablation):** What is the contribution of (a) failure-triple filtering and (b) embedding-based matching to VC quality?
- **RQ4 (LLM Sensitivity):** How stable are the channel-level results across different LLMs from multiple model families?

TABLE II

DATASET AND PER-CONDITION VC FLOW. CH-A AND CH-B COVER DISJOINT SYRS SUBSETS ($208 + 142 = 350$) AND ABSTAIN VIA THE `NO_NOVEL_VC_FOUND` SENTINEL; "CAND." COUNTS CANDIDATE VCS (AN ITEM MAY YIELD MORE THAN ONE), AND THE CAND. → EVAL. GAP REMOVES DUPLICATE OR MALFORMED CANDIDATES BEFORE REVIEW. THE BASELINE APPLIES THE SAME SENTINEL OVER ALL 350 ITEMS. SOURCES: 696 GITHUB ISSUES, 364 NHTSA COMPLAINTS.

Condition	Items	Abst.	Cand.	Eval.
A (GitHub)	208	121	88	87
B (NHTSA)	142	13	131	128
Baseline (all SYRS)	350	73	277	274

Evaluation totals $87 + 128 + 274 = 489$ VCs spanning 307 unique SYRS; 43 items yielded no evaluated VC. Ch-A and Ch-B share no SYRS; the baseline spans all 350 items (overlapping both channels).

B. Dataset

a) SYRS corpus.: We apply the pipeline to 350 unique automotive ECU SYRS items drawn from a proprietary specification corpus of 675 items, covering body control (ESCL, EPB, SCCM), communication stack (UDS, CAN), diagnostics, and system state management components. The channel mapping is disjoint by construction: Ch-A covers 208 communication-stack and diagnostics items; Ch-B covers 142 body-control, HMI, and functional-safety items. All items follow the ASPICE SYS.2 format with explicit actor, trigger, quantitative boundary, and expected behaviour, and each carries human-annotated golden VCs.

b) Generated VCs.: Across both channels and the no-context baseline, 489 VCs entered expert evaluation (Table II):

C. Evaluation Metrics

a) Accept rate.: Fraction of VCs whose majority-vote label is Accept (Framework B: $C \geq 3$ AND $N = \text{Y}$ AND $U = \text{Y}$).

b) Wilson 95 % CI.: Confidence interval for the accept rate using the Wilson score method [36].

c) Fisher's exact test + Holm–Bonferroni correction.: Each channel is tested against the baseline with Fisher's exact test [37]; p-values are corrected with Holm–Bonferroni [38] across three simultaneous comparisons (Ch-A, Ch-B, and A+B combined vs. baseline).

d) Cohen's h.: Effect size for proportion comparisons [39]: $h \geq 0.2$ small; $h \geq 0.5$ medium; $h \geq 0.8$ large.

e) Generation-rate proxy.: When the LLM determines that the retrieved failure context offers no test angle not already addressed by existing golden VCs, it is prompted to abstain, emitting the sentinel string `NO_NOVEL_VC_FOUND` instead of a VC. The *generation rate* is the fraction of SYRS items for which the model generates a VC (i.e., does *not* abstain); a high generation rate under active filtering indicates the model is finding genuinely new failure angles, while near-universal generation (>93 %) signals the model is ignoring the abstention instruction. We use the generation rate as a lightweight quality proxy for LLM sensitivity runs where full human evaluation is not available (RQ4).

f) Abstention rates.: In the primary evaluation, Ch-A (GitHub) abstains on 121 of 208 SYRS items (58.2 %); Ch-B (NHTSA) abstains on 13 of 142 items (9.2 %). The large difference reflects the vocabulary alignment gap: NHTSA complaint records, once filtered into structured fault modes, map tightly to the component-level language of body-control and chassis SYRS items, while GitHub issue vocabulary is more varied relative to communication-stack and diagnostics items, triggering abstention more often. This asymmetry is itself a qualitative finding: the pipeline selectively generates VCs rather than producing output indiscriminately.

g) Matching precision.: To establish confidence in the upstream retrieval quality, we manually evaluated a stratified random sample of 200 requirement–evidence pairs (100 per channel, seed = 42, similarity range 0.45–0.78). A pair is labelled *relevant* if the retrieved evidence describes a failure mode or boundary condition observable in the ECU under test (not merely a topically adjacent software defect or host-toolchain issue). The pipeline achieves an overall matching precision of **73.5 %** (Ch-A GitHub: **61.0 %**; Ch-B NHTSA: **86.0 %**), confirming that the evidence reaching the generator is failure-relevant.

D. Inter-Annotator Agreement

a) Rater profiles.: We recruited four raters to span the three stakeholder roles in ASPICE V&V: process ownership, academic methodology, and hands-on practice. All have direct automotive software engineering experience; their disciplinary mix (two industry practitioners, one process/safety specialist, and one academic researcher) guards against single-perspective bias in a multi-criterion framework. **E1** (Embedded Systems Project Manager, 10 yr industrial experience) acts as adjudicator, resolving split votes using defined criteria. **E2** (ASPICE assessor and functional-safety consultant) contributes direct SYS.2 process authority. **E3** (PhD candidate in automotive software testing) provides systematic, methodology-grounded judgement. **E4** (Senior functional-test engineer, automotive OEM/Tier-1) anchors evaluations in physical V&V executability. This industry–academia panel structure replicates the design of CrUISE-AC [3].

b) Main evaluation (RQ1/RQ2).: E2, E3, and E4 independently scored the 489 VCs, each VC receiving exactly two independent external ratings; E1 (adjudicator) resolved split decisions under Framework B. The accept/reject label is the majority vote among the raters who scored each VC. We use Gwet's AC1 [40], [41] rather than Cohen's κ to avoid the kappa paradox [42] that arises from the high prevalence of the Accept class under Framework B. Pairwise AC1 is reported on the Completeness criterion ($C \geq 3$).

c) Ablation evaluation (RQ3).: The 50-item subset comprises the first 50 Ch-A items in corpus iteration order; all are Communication-Stack requirements. This homogeneity strengthens internal comparability across ablation conditions (category cannot confound the contrast) but restricts RQ3's external scope to that category (§VI). All four raters (E1–E4) independently score the 50-item Ch-A ablation subset;

TABLE III
RQ1: ACCEPT RATES VS. BASELINE (THREE-CRITERION PROTOCOL; MAJORITY VOTE)

Condition	VCs	Accept	Rate	Holm-p
Baseline (no-context)	274	124	45.3 %	—
Channel A (GitHub)	87	57	65.5 %	0.0007***
Channel B (NHTSA)	128	86	67.2 %	<0.001***
A+B combined	215	143	66.5 %	<0.001***

*** $p < 0.001$ (Holm–Bonferroni corrected).

the accept/reject label is the majority vote of E2, E3, and E4 ($\geq$2/3) with E1 as adjudicator (matching the main-evaluation protocol), and all four raters' scores are used for the pairwise AC1 analysis.

E. Baselines

a) No-context baseline.: Qwen-Max prompted with SYRS text only (no failure context). This represents the zero-shot VC generation scenario and is the primary comparison point for RQ1.

b) CrUISE-AC [3].: The closest published system, targeting acceptance criteria for user stories from crowd-reported bugs. Accept rates and IAA values are compared in §V-A to contextualise our results.

F. Computational Cost

The core pipeline—Qwen-Plus filtering, Qwen-Max generation across both channels, and the ablation conditions—cost approximately **US$3** in API fees. The seven-model RQ4 sensitivity sweep added roughly US$12 more, dominated by reasoning-heavy models whose long completions account for most output tokens, for a total on the order of US$15. Exact figures depend on time-varying provider pricing and are reported approximately. Embedding-based matching was negligible ($<$ \$0.05); the Domain-2 pilot (85 Diagnostics SYRS, Ch-A only) added approximately \$0.5.

V. RESULTS AND ANALYSIS

A. RQ1: Effectiveness of Failure-Grounded Prompting

Table III and Fig. 3 summarise acceptance rates for all conditions. Channels A and B both significantly outperform the baseline after Holm–Bonferroni correction. Channel A (GitHub) achieves 65.5 % acceptance (57/87), OR = 2.30, Cohen's $h = 0.41$, Holm-$p = 0.0007$. Channel B (NHTSA) achieves 67.2 % (86/128), OR = 2.48, $h = 0.45$, Holm-$p < 0.001$. The combined A+B pipeline reaches 66.5 % (143/215), confirming both channels contribute; the joint A+B vs. baseline Fisher test yields $p < 0.001$, OR = 2.40, $h = 0.43$.

a) IAA.: Pairwise Gwet's AC1 on the Completeness criterion ($C \geq 3$) is 0.84 (E2–E3; 86 % agree), 0.65 (E2–E4; 74 % agree), and 0.69 (E3–E4; 76 % agree), averaging 0.73. The two lower pairs both involve E4, reflecting genuine disciplinary divergence between E4's physical-executability perspective and the requirements/process perspectives of E2 and E3, a discipline-level finding documented in §VI. The mean AC1 is comparable to CrUISE-AC's reported AC1 = 0.74 [3], noting that our three-criterion protocol imposes a strictly stronger acceptance predicate than CrUISE-AC's binary acceptance judgement.

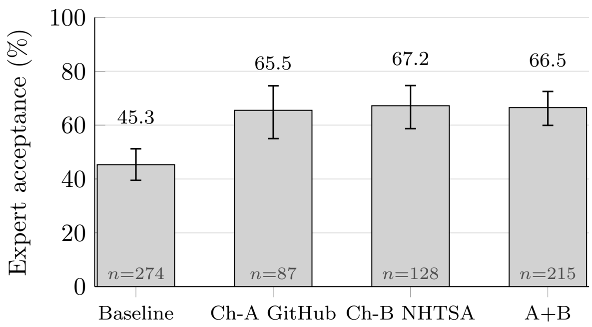

Fig. 3. RQ1: expert acceptance by condition with Wilson 95 % CIs (three-criterion protocol, majority vote). Both channels significantly outperform the no-context baseline (Holm-corrected $p < 0.001$).

b) Per-item accepted-VC yield.: Conditional acceptance rates compare *generated* VCs and are therefore not commensurable across conditions with different abstention rates. A common-denominator metric is the *per-item yield*: accepted VCs divided by the total items assigned to that channel/condition.

TABLE IV
PER-ITEM ACCEPTED-VC YIELD (COMMON-DENOMINATOR METRIC).

Condition	Items	Gen. rate	Accept rate	Yield
Baseline	350	79.1 %	45.3 %	0.35
Ch-A (GitHub)	208	41.8 %	65.5 %	**0.27**
Ch-B (NHTSA)	142	90.8 %	67.2 %	**0.61**

Ch-A's per-item yield (0.27) is below the baseline (0.35): the pipeline generates a VC for only 41.8 % of its items, though 65.5 % of those are accepted. Ch-A is therefore a *precision channel* (selective generation at high quality) rather than a high-recall supplement to the baseline; engineers who need broad VC coverage over a corpus can combine Ch-A with the baseline. Ch-B's yield (0.61) substantially exceeds the baseline, indicating that complaint-grounded VCs cover most assigned items and are accepted at a higher rate.

Answer to RQ1: Failure-grounded prompting with GitHub issues (Ch-A) and NHTSA complaints (Ch-B) significantly outperforms a no-context baseline: +20 pp and +22 pp respectively (Holm-corrected $p < 0.001$, $h = 0.41$–0.45, small-to-medium effect). IAA (mean Gwet's AC1 = 0.73) is comparable to prior ICSE work under a stricter evaluation criterion. Per-item yield: Ch-B = 0.61, Baseline = 0.35, Ch-A = 0.27; Ch-A is a high-precision selective channel, not a high-yield one.

B. RQ2: Channel–Domain Fit

Because the two channels are mapped to disjoint requirement subdomains, RQ2 concerns where each channel fits

TABLE V
SCOPE-STRATIFIED ACCEPTANCE BY ECU CATEGORY (THREE-CRITERION PROTOCOL, MAJORITY VOTE). "—" = NOT IN BASELINE CORPUS.

Category	Ch.	Pipeline		Baseline	
		Acc/N	Rate	Acc/N	Rate
Body-Control & Chassis (Ch-B)					
IO / HMI	B	23/31	74.2 %	18/39	46.2 %
Functional Safety	B	23/32	71.9 %	—	—
Body Control	B	28/41	68.3 %	21/42	50.0 %
Comm. Stack & State Mgmt (Ch-A/B)					
Communication Stack	A	19/31	61.3 %	31/55	56.4 %
System State Mgmt	B	12/24	50.0 %	11/28	39.3 %
Diagnostics (Ch-A)					
Diagnostics (DCM)	A	29/44	65.9 %	25/64	39.1 %
Diagnostics (DEM)	A	4/6	66.7 %	14/33	42.4 %
Diagnostic Services	A	5/6	83.3 %	4/13	30.8 %

rather than which channel performs better in a head-to-head comparison. Within its assigned domain, each channel yields a category-dependent uplift over the no-context baseline averaging +20 pp (Channel A +20.7 pp, Channel B +21.3 pp; category-matched,[3] derived from Table V), each supported by an independent small-to-medium effect size relative to the baseline (Channel A $h = 0.41$; Channel B $h = 0.45$). The grounding mechanism differs by source: NHTSA complaints are consumer-authored narratives tagged with component metadata, and once Stage-2 filtering removes their colloquial noise the residual component-level fault modes align closely with SYRS vocabulary; GitHub issues offer higher volume (696 vs. 364 records) at the cost of greater terminological variance.

a) Scope-stratified acceptance.: Table V breaks down acceptance by ECU component category. Body-control and chassis items (covered by Ch-B) show consistently high acceptance: IO/HMI $=$ $74.2\,\%$ ($+28$ pp), Functional Safety $=$ $71.9\,\%$, and Body Control $=$ $68.3\,\%$: filtered NHTSA complaint narratives map tightly to sensor-level and state-machine specifications in these categories. Communication-stack items (Ch-A) improve more modestly ($61.3\,\%$ vs. $56.4\,\%$; $+4.9$ pp), reflecting greater vocabulary variation in GitHub issues for protocol-specific SYRS items.

[3]*Pooled* deltas (the +20/+22 pp reported in RQ1) compare each channel against the full no-context baseline, whereas *category-matched* deltas (RQ2) compare it against the baseline restricted to that channel's own requirement categories.

Answer to RQ2: Within each channel's assigned domain: Ch-B (NHTSA) achieves 67.2 % acceptance on body-chassis requirements; Ch-A (GitHub) achieves 65.5 % on communication-stack and diagnostics requirements. Channel and requirement category are disjoint by design, so the results characterise per-channel performance within-domain rather than comparing channels head-to-head. Within Ch-B's domain, body-control and chassis items reach 68–74 %, where filtered complaint narratives align tightly with sensor-level and state-machine SYRS vocabulary. Within Ch-A's domain, communication-stack items improve modestly (+4.9 pp vs. baseline), reflecting greater vocabulary variation in GitHub issues for protocol-specific SYRS items.

Table I (§III) shows one accepted VC from each channel, illustrating their complementary roles.

C. RQ3: Ablation

Table VI reports human acceptance rates on the 50-item Ch-A evaluation subset under the three ablation conditions.

TABLE VI
RQ3: HUMAN ACCEPTANCE RATE ON 50-ITEM CH-A SUBSET (4-RATER MAJORITY VOTE, C $\geq$ 3 AND N=Y AND U=Y).

Condition	Accepted	Rate
Full Field2VC (FL, filter + embed)	29/50	58.0 %
Abl-LM (LLM-Match*)	29/50	58.0 %
Abl-NF (No Filter, embed)	41/50	82.0 %

*Replicates CrUISE-AC matching strategy [3]: LLM binary relevance classification, no triple filtering.

a) Embedding matching vs. LLM binary classification.: With all three conditions evaluated on the same 50-item subset, the clean matching-method comparison is between the two *unfiltered* ablations: Abl-NF (embedding on raw text, no filtering) achieves 82.0 % versus 58.0 % for Abl-LM (LLM binary relevance classifier, equivalent to CrUISE-AC's matching strategy [3]), a 24 pp gap attributable to the retrieval mechanism since filtering is absent in both. The full pipeline (FL, filter + embed) matches Abl-LM exactly at 58.0 %: when filtering is active, the accept rate is determined by the Stage-2 abstention decision, not by the matching step.

b) Why filtering is necessary at scale.: Although Abl-NF achieves the highest human accept rate on the subset, its generation-rate proxy reaches 96–100 % across all channels. Without the $\langle$Trigger|Protocol Behavior|Fault Mode$\rangle$ filtering triple, raw embedding matches semantically adjacent but thematically misaligned issues to SYRS items, causing the generator to produce a VC for nearly every SYRS regardless of genuine relevance. For Ch-A, the pipeline generates VCs for 87 of 208 SYRS items (41.8 % generation rate); among those 87 VCs the human accept rate is 65.5 %: the pipeline abstains when the failure evidence does not support a grounded VC. The generation-rate proxy is meaningful only when filtering is active (§V-D).

c) IAA (FL condition).: On the 50-item FL evaluation, individual accept rates were: E1 33/50 (66.0 %), E2 24/50 (48.0 %), E3 47/50 (94.0 %), and E4 27/50 (54.0 %), yielding a majority vote (E2+E3+E4 $\geq$ 2/3) of 29/50 (58.0 %). Six rows produced E1–MV disputes: five where E1 accepted but the MV rejected (Extended-Addressing scope mismatch or tool-layer contamination), and one (Row 013) where E1 rejected but the MV accepted (BufferedReader ECU wake-up transition boundary). MV prevailed in all six cases per the pre-defined adjudication protocol. Mean Gwet's AC1 across 4 raters is 0.63 for Abl-NF (substantial agreement) and 0.37 for Abl-LM (moderate agreement), corroborating the quality difference: experts disagree more on LLM-classifier-matched VCs. Pairwise percentage agreement: Abl-NF ranges from 66 % (E1–E4) to 92 % (E1–E3); Abl-LM spans 34 % (E3–E4) to 88 % (E1–E2), with E3–E4 divergence driven by systematic methodological differences on Extended-Addressing conditions (§VI).

TABLE VII
REJECTION ATTRIBUTION FOR THE 21 MV-REJECT ITEMS (RQ3, FL CONDITION, $n = 50$)

Category	Items	% Rej.	% of 50
Tool contamination (Stage-2 DUT-boundary leakage)	14	67 %	28 %
Semantic drift (VC–req. scope mismatch)	5	24 %	10 %
Generation failure (timeout / empty output)	2	10 %	4 %
Total	**21**	**100 %**	**42 %**

Answer to RQ3: On the 50-item ablation subset, FL (full pipeline) and Abl-LM both accept 58.0 % (29/50); Abl-NF (embed, no filter) accepts 82.0 % (41/50). The clean matching-method contrast is Abl-NF vs. Abl-LM (both unfiltered): 82.0 % vs. 58.0 %, a 24 pp embedding advantage attributable to the retrieval mechanism since filtering is absent in both. When filtering is active (FL), the matching method is not the dominant variable: FL and Abl-LM converge at 58.0 %. Filtering is necessary at scale: without it, generation rates reach 96–100 % indicating ungrounded generation. Of the 21 FL rejects, 14 (67 %) arise from Stage-2 DUT-boundary leakage (host-layer tool properties matched to ECU requirements), 5 (24 %) from semantic drift, and 2 (10 %) from generation failure (Table VII). IAA: mean Gwet's AC1 = 0.63 (Abl-NF) / 0.37 (Abl-LM).

D. RQ4: LLM Sensitivity

a) Sentinel adherence divides model families.: Five of seven models generate VCs for >93 % of SYRS items regardless of knowledge-source relevance, effectively ignoring the `NO_NOVEL_VC_FOUND` sentinel. Qwen-Max and Qwen-Plus are the only models that emit the sentinel discriminatively: Qwen-Max (sensitivity) yields 40.4 % for Channel A (where GitHub vocabulary is most varied), rising to 93.7 % for Channel B; Qwen-Plus shows a similar but shallower gradient (59.6 %, 94.4 %). This sentinel-adherence difference is a practical finding for practitioners: the generation-rate proxy is only a reliable quality signal when the underlying LLM respects the generation-suppression instruction.

TABLE VIII
RQ4: LLM SENSITIVITY. THE PRIMARY RUN IS HUMAN-EVALUATED (ACCEPT RATE); ALL SENSITIVITY RUNS REPORT THE GENERATION-RATE PROXY ONLY, WHICH IS NOT DIRECTLY COMPARABLE TO THE ACCEPT RATE.

LLM	Ch-A	Ch-B
Primary run — human accept rate		
Qwen-Max	65.5 %	67.2 %
Sensitivity runs — generation-rate proxy		
Sentinel-adherent		
Qwen-Max (sensitivity)	40.4 %	93.7 %
Qwen-Plus	59.6 %	94.4 %
Non-sentinel-adherent (>93 % on both channels)		
DeepSeek-v4-Pro	99.0 %	100.0 %
DeepSeek-v4-Flash	100.0 %	99.3 %
Kimi-K2.6	94.2 %	97.2 %
GLM-5	98.6 %	98.6 %
Qwen3.6-Plus	100.0 %	99.3 %

Accept rate = human-evaluated main run (Table III); generation-rate proxy rows have no human evaluation. The two metrics are not directly comparable.

b) Channel ordering is knowledge-source-driven, not LLM-driven.: Across both sentinel-adherent (Qwen-family) models, the B > A quality ordering holds, consistent with the scope-stratified analysis in RQ2.

Answer to RQ4: Across 7 LLMs, sentinel adherence divides model families: only Qwen-Max and Qwen-Plus emit `NO_NOVEL_VC_FOUND` discriminatively, making the generation-rate proxy meaningful only for these models. Qwen-Max (sensitivity): ChA 40.4 %, ChB 93.7 %; Qwen-Plus: ChA 59.6 %, ChB 94.4 %. The B > A quality ordering holds across both sentinel-adherent (Qwen-family) models, consistent with a knowledge-source-driven rather than LLM-driven pattern.

VI. DISCUSSION AND THREATS TO VALIDITY

A. Failure Mode Analysis

Systematic analysis of the 72 rejected VCs across Channels A and B reveals three distinct failure modes that account for the gap between channel acceptance rates.

FM-1: Boundary Underspecification (the most common mode). The generated VC is structurally well-formed but lacks quantitative parameters (timing constraints, byte constants, threshold expressions). Root cause: source issues describe qualitative fault behaviour without numeric bounds; the LLM mimics the issue vocabulary and omits boundaries that appear only in the SYRS text. More common in Channel A; less prevalent in Channel B, where the component metadata and filtered fault modes of complaint records constrain the generated quantitative bounds. *Recommendation:* Inject all numeric literals from the SYRS text as mandatory constraints in the generation prompt; flag generated VCs that reference none of them.

FM-2: Scenario Transplant. The VC correctly identifies the right ECU component and fault type but embeds the test procedure in an irrelevant consumer-driving scenario lifted from an NHTSA complaint narrative (e.g., "drive the vehicle for 30 minutes with slight left/right steering inputs"), making it operationally impossible in a standard HIL bench. This is the dominant Channel-B mode. *Recommendation:* Apply a scenario-abstraction post-processing step that strips road-context sentences and substitutes laboratory-equivalent trigger stimuli.

FM-3: Filtering Collapse (thematic misalignment at scale). Without Stage 2 filtering, embedding similarity alone retrieves topically adjacent but thematically misaligned issues, and the LLM generates syntactically coherent VCs that test conditions unrelated to the SYRS under test (e.g., a UDS service-0x29 test generated for a framing requirement). This mode operates *at corpus scale*, not on the 50-item human-evaluation subset: across the full corpus Abl-NF's generation rate reaches 96–100 % (§V-C), whereas the filtering-enabled full pipeline abstains on the majority of items (58.2 % of Ch-A SYRS, generating for only 41.8 %). The gap between near-universal generation and selective abstention is the volume of thematically misaligned matches that filtering suppresses; on the small 50-item ablation subset, by contrast, Abl-NF's human accept rate is comparatively high, so the two settings describe different populations. The full pipeline suppresses this mode via filtering-enabled abstention. *Recommendation:* The filtering stage is not optional for safety-critical applications; at minimum, enforce an AUTOSAR module tag guard before embedding matching.

B. Threats to Validity

a) Construct validity.: The accept/reject label depends on human raters applying Framework B ($C \geq 3$ AND $N = \mathrm{Y}$ AND $U = \mathrm{Y}$). To mitigate subjectivity, we recruited four raters—three independent scorers (E2–E4) and an adjudicator (E1)—with automotive RE backgrounds, and computed Gwet's AC1 [40] to account for the kappa paradox [42] arising from high Accept-class prevalence. The adjudicator (E1) was not blind to the generation channels (Channel A/B/Baseline) and is an author of this paper, which introduces a potential risk of confirmation bias—particularly for the high acceptance rate of Channel B (67.2 %), which relied heavily on E1's adjudication. To mitigate this, E1's rulings were strictly anchored to the pre-registered Framework B, and we tracked peer-rater alignment: all 58 disputed Channel-B acceptances were cross-verified to follow the rulings of E2 (an independent ASPICE assessor), preventing arbitrary author rulings. Beyond these safeguards, we report the headline rates under two alternative adjudication assumptions rather than claim independence (E1 resolved split decisions on 50/87 Channel-A, 89/128 Channel-B, and 146/274 baseline items). Under a conservative *no-E1 floor* that treats every split as a reject, Channel A falls to 42.5 % against a 43.4 % baseline under the same rule, and Channel B to 15.6 %: the per-channel uplift does not survive this worst case, so the headline effect is partly adjudication-dependent. Restricted instead to the *agreed subset*—items both external raters labelled identically without adjudication—Channel A is accepted on all 37/37 items (100 %) versus 119/128 (93.0 %) for the baseline, so the Channel-A effect persists without adjudication on clear-cut cases (a self-selected subset); Channel B accepts only 51.3 % of its agreed subset, confirming that the NHTSA channel leans more heavily on adjudication than the GitHub channel. Single-adjudicator, non-blind scoring thus remains a genuine limitation: unlike CrUISE-AC [3], which accepts on a majority of $\geq$3-of-4 *independent* experts without an adjudicator, our protocol pairs two external raters with an adjudicator, and broadening to a larger blinded, independent panel under majority vote is the natural strengthening. The E2–E4 and E3–E4 Completeness pairs (0.65 and 0.69) are the lowest of the three and both involve E4, reflecting genuine disciplinary divergence between E4's physical-executability perspective and the requirements/process perspectives of E2 and E3, documented as a discipline-level finding rather than a measurement failure. The generation-rate proxy (RQ4) approximates VC novelty without human evaluation but cannot distinguish high-quality from low-quality novel VCs; proxy validity is confirmed only when filtering is active (§V-C).

b) Evaluator methodology divergence (Abl-LM).: The Abl-LM condition exhibits a markedly low pairwise agreement between E3 and E4 (34 %, $\mathrm{AC1} = -0.24$), reflecting a systematic methodological difference rather than random disagreement. E3 applied a *protocol-mechanism relevance* criterion grounded in ISO 15765-2: since N_Bs/N_Cs timers and STmin parameters decouple from addressing mode, VCs that specify Extended Addressing (ExtAddr) as a test precondition are judged on the merit of their protocol-level content, treating the ExtAddr clause as a non-modifying context. E2 and E4 applied a *deployment-readiness* criterion: any ExtAddr condition that is inconsistent with a physical-addressing test environment renders the entire VC non-deployable, warranting rejection. Both perspectives reflect legitimate industry practice. The majority-vote mechanism ($\geq$2/3 of E2, E3, E4) ensures that the more conservative E2+E4 position prevails on contested items, making the reported 58.0 % Abl-LM acceptance rate a conservative lower bound.

c) Internal validity.: The Fetch stage uses repositories chosen for ECU-adjacent vocabulary; different choices might alter results. We mitigate by drawing from three distinct stacks (UDS diagnostics, CAN transport, ECU toolchain). A further internal threat is a small prompt-budget asymmetry: the golden-VC context passed to the generator is truncated at 800 characters in the pipeline conditions but 600 characters in the baseline; we disclose this asymmetry rather than re-run, as it affects only the do-not-repeat context, not the failure evidence under study. The absence of a hard similarity gate (§III-C) is supported post-hoc: every VC that survived sentinel abstention had a top-1 match similarity of at least 0.50, and above this empirical floor similarity does not predict expert acceptance (quartile accept rates 57–76 %, non-monotonic); a hypothetical 0.60 gate would have discarded VCs with a *higher* accept rate (70.0 %) than those retained (63.5 %). The Channel-B trace in Table I makes this concrete. Its 0.60 top-1 similarity sits at the very boundary such a gate would cut, and the retrieved complaint reports low-beam flashing triggered by high-beam activation and ignition—an *internal electrical* perturbation—rather than the *external optical* AC-flicker stimulus SYRS.0177 specifies. Despite this mechanism divergence, Stage 4—conditioned on the SYRS text and its golden VC, not on the evidence alone—recovered the shared verification concern (low-beam ON-state stability under perturbation) and produced a VC unanimously accepted by all assigned raters. We read this as architectural, not coincidental: gate-free retrieval (§III-C) supplies coarse candidate evidence, while requirement-grounded generation keeps the resulting VC anchored to the SYRS intent even when the evidence mechanism diverges. It is a scope condition, not a guarantee—such recovery is expected only where the requirement states its intent precisely enough to anchor generation, as SYRS.0177's quantitative bounds ($<$1000 lux, $>$4 s) do.

d) Evidence grounding vs. abstention.: One might ask whether the acceptance gains reflect the failure evidence or merely the Stage-2 abstention mechanism, which withholds output on low-confidence items and could inflate the accept rate by reporting only a favourable subset. We separate the two using per-item yield (accepted VCs over *all* assigned items, Table IV), which charges every abstention as a miss. Channel B attains its category-matched $+21.3$ pp uplift while abstaining on only $9.2\,\%$ of items ($129/142$ generated), so its improvement cannot be attributed to selective silence. Channel A behaves differently: it abstains on $58.2\,\%$ of items and its yield (0.27) falls below the baseline (0.35)—it trades recall for precision rather than uniformly improving generation. We therefore claim that failure grounding improves VC quality where filtered evidence aligns with requirement vocabulary (Channel B), not that it uniformly raises generation across all unselected requirements; an abstention-matched head-to-head isolating the marginal effect of evidence is left to future work.

e) DUT-boundary filtering failure (RQ3).: Stage 2 extracts ⟨Trigger | Protocol Behavior | Fault Mode⟩ triples and abstains when no grounded triple can be formed. In the 50-item FL evaluation, 14 of the 21 MV-Reject VCs (67 %) arose from DUT-boundary leakage: python-can host-layer properties—log levels, file-descriptor limits, Windows-specific socket IDs, periodic sender timing bugs (hardbyte/python-can#888)—were extracted as ECU-observable behaviours and passed the triple filter. The root cause is that issue reports mix host-side and device-side concerns within the same thread; without an explicit component-layer annotation, Stage 2 cannot reliably infer which side of the DUT boundary a property belongs to. The resulting VCs ask the ECU to verify conditions that only the test harness can observe, rendering them non-deployable on a standard HIL bench. As a concrete counterexample, for SYRS.0693_A the generator produced a well-formed VC asserting correct behaviour at file-descriptor counts beyond 1023 and at specific host log levels—`python-can` library properties, not ECU-observable behaviour—which all but one rater rejected; such discriminated rejections confirm that the evaluation is not merely rubber-stamping plausible-looking output. A guard that checks for known test-tool package namespaces (e.g., `python-can`, `udsoncan`, `isotp`) before triple extraction would suppress the majority of these false positives.

f) External validity.: The primary evaluation covers Channel A's communication-stack and diagnostics requirements and Channel B's body-control, HMI, and functional-safety requirements, all from one proprietary project. To probe further generalisability, we applied Channel A (Qwen-Max) to a separate Domain-2 pilot of 85 Diagnostics ECU SYRS items (DCM, DEM, Diagnostic Services). The pilot produced 15 novel VCs (17.6 % generation-rate proxy; mean top-1 similarity 0.625) with domain-appropriate content. We subsequently conducted a full three-rater evaluation (E2, E3, E4 from the primary study) of all 15 VCs under Framework B. **13 of 15 VCs were accepted** (86.7 %, Wilson 95 % CI: 62–96 %), exceeding the primary Channel-A acceptance rate (65.5 %). Inter-rater agreement was substantial for Novelty (mean Gwet's AC1 = 0.69) and moderate for Usability (mean AC1 = 0.42); the lower Usability agreement reflects E4's stricter physical-executability criterion (e.g., rejecting DoIP-based VCs on CAN-only benches), while E2 and E3 agreed perfectly (AC1 = 1.00 on both dimensions). The lower generation rate vs. the body-control domain (17.6 % vs. 90.8 % for Ch-B overall: 129/142 SYRS) reflects that Diagnostics SYRS items are more self-contained: the pipeline abstains more often because UDS/DTC failure modes in GitHub issues are less tightly aligned with individual Diagnostics SYRS items than NHTSA complaint profiles are with body-control SYRS items; the higher acceptance rate among generated VCs (86.7 %, $n = 15$; Wilson CI 62–96 %, overlapping the primary rate) provides preliminary evidence that cross-domain transfer can succeed without domain-specific prompt engineering. A second generalisation limit concerns models: the B $>$ A channel ordering is established only on the two sentinel-adherent models, both from the Qwen family, so its stability across other model families is untested. Generalisability to powertrain, ADAS, or gateway ECU domains remains unexamined; NHTSA effectiveness may not transfer to domains without a comparable regulator-operated complaint database. The RQ3 ablation subset is additionally homogeneous: all 50 items are Communication-Stack requirements (the first 50 in corpus iteration order), so the ablation conclusions are internally well-controlled but do not automatically extend to other ECU categories.

g) Conclusion validity.: We apply Holm–Bonferroni correction [38] across three simultaneous Fisher's exact tests to control family-wise error rate at $\alpha = 0.05$. Wilson 95 % CIs [36] are reported for all acceptance rates. Cohen's h [39] is reported alongside p-values to prevent over-interpretation in small-to-medium samples. Several scope-stratified cells in Table V are small (Diagnostics DEM and Diagnostic Services each n=6); their per-category rates carry wide Wilson intervals and are reported as descriptive breakdowns rather than independently powered comparisons.

VII. Conclusion

We presented Field2VC, a five-stage pipeline that completes automotive ECU verification criteria by grounding LLM generation in empirical failure evidence from two complementary external channels: GitHub issues (Ch-A) and NHTSA consumer complaints (Ch-B). A four-expert evaluation (three independent scorers and one author-adjudicator) of 489 generated VCs spanning 307 unique ASPICE SYS.2 items yields four findings:

RQ1. Failure-grounded prompting with Ch-A and Ch-B achieves 65.5 %–67.2 % expert acceptance versus a 45.3 % no-context baseline (Holm-corrected $p < 0.001$, $h = 0.41$–0.45, IAA: mean Gwet's AC1 = 0.73).

RQ2. Within their assigned subdomains, both sources are effective: each yields a category-dependent uplift over the no-context baseline (Channel A +20.7 pp, Channel B +21.3 pp; category-matched), supported by an independent small-to-medium effect size ($h = 0.41$ and $h = 0.45$, respectively). NHTSA's component-tagged field-failure descriptions, once filtered into structured fault modes, align closely with SYS.2 vocabulary; scope-stratified analysis shows the strongest gains in body-control and chassis categories (68–74 %) and more modest improvement in communication-stack items (61.3 %).

RQ3. On the 50-item ablation subset, FL (full pipeline) and Abl-LM both accept 58.0 % (29/50); the clean embedding vs. LLM-match comparison is Abl-NF (82.0 %) vs. Abl-LM (58.0 %), a 24 pp gap attributable to matching method since filtering is absent in both. When filtering is active, acceptance is driven by the Stage-2 abstention decision, not the retrieval step. Filtering (Stage 2) is necessary at scale: without it, near-universal generation rates (96–100 %) indicate ungrounded matching.

RQ4. The B > A channel quality ordering holds across both sentinel-adherent (Qwen-family) models, consistent with a knowledge-source-driven rather than LLM-driven pattern; five of seven models fail to respect the generation-suppression sentinel (>93 % generation rate), making sentinel adherence a prerequisite for using the generation rate as a quality proxy in multi-LLM pipelines.

A pilot on 85 Diagnostics ECU SYRS (Domain 2, UDS/DTC subdomain) offers preliminary evidence of cross-domain generalisability: all 15 novel VCs from Channel A received full three-rater evaluation (E2/E3/E4), with 13 of 15 (86.7 %) accepted under Framework B, exceeding the primary Channel-A rate (65.5 %) without any domain-specific prompt engineering.

Two reusable lessons emerge: (1) *the usefulness of a failure knowledge source depends on whether, after filtering, its vocabulary aligns with the quantitative, component-level language that safety-critical requirements demand*: regulator-collected complaint records, once filtered into structured fault modes, satisfy this more consistently than community-sourced issue reports; and (2) *generation-rate proxy validity depends on the LLM's adherence to generation-refusal instructions*, which must be validated before use as a quality signal.

To support replication, we release the full de-identified evaluation dataset (the SYRS corpus covering all 307 evaluated items with their expert-annotated golden VCs, all 489 generated VCs with per-rater labels and adjudication, the Framework B protocol, the LLM prompts, and the analysis scripts) as a reproducible benchmark for VC-completion research (Data Availability).

a) Future Work.: Automated VC formalisation into machine-checkable predicates for model-based testing, a traceability matrix linking each accepted VC to its source failure record, and replication on powertrain and ADAS domains are planned as direct extensions of this work.

References

[1] B. Boehm and V. R. Basili, "Software defect reduction top 10 list," *Computer*, vol. 34, no. 1, pp. 135–137, 2001.

[2] *Automotive SPICE Process Reference and Assessment Model, Version 4.0*, VDA Quality Management Center, 2023.

[3] S. Schwedt and T. Ströder, "From Bugs to Benefits: Improving User Stories by Leveraging Crowd Knowledge with CrUISE-AC," in *47th IEEE/ACM International Conference on Software Engineering, ICSE 2025, Ottawa, ON, Canada, April 26 - May 6, 2025*. IEEE, 2025, pp. 1385–1395.

[4] T. Ward, "Areas of Improvement for Autonomous Vehicles: A Machine Learning Analysis of Disengagement Reports," 2024, arXiv:2408.00051.

[5] J. R. Palit, V. K. Kumarasamy, and O. A. Osman, "Data-Driven Analysis of Crash Patterns in SAE Level 2 and Level 4 Automated Vehicles Using K-means Clustering and Association Rule Mining," 2026, arXiv:2512.22589.

[6] A. Mavin, P. Wilkinson, A. Harwood, and M. Novak, "Easy approach to requirements syntax (EARS)," in *Proc. 17th IEEE Int. Requirements Engineering Conf. (RE)*, 2009, pp. 317–322.

[7] D. Luitel, S. Hassani, and M. Sabetzadeh, "Improving Requirements Completeness: Automated Assistance through Large Language Models," 2024, arXiv:2308.03784.

[8] Z. Wu, X. Chen, Z. Jin, M. Hu, and D. Jin, "Unlocking the silent needs: Business-logic-driven iterative requirements auto-completion," in *47th IEEE/ACM International Conference on Software Engineering, ICSE 2026, Rio de Janeiro, Brazil, April 14 - 20, 2026*. ACM/IEEE, 2026.

[9] X. Lian, S. Wang, H. Zou, F. Liu, J. Wu, and L. Zhang, "Incorporating verification standards for security requirements generation from functional specifications," in *Proc. ACM SIGSOFT Int. Symp. Foundations of Software Engineering (FSE)*, 2025.

[10] F. Niu, R. Pan, L. C. Briand, H. Hu, and K. Koravadi, "TVR: Automotive System Requirement Traceability Validation and Recovery Through Retrieval-Augmented Generation," *CoRR*, 2025, arXiv:2504.15427.

[11] A. Masoudifard, M. M. Sorond, M. Madadi, M. Sabokrou, and E. Habibi, "Leveraging Graph-RAG and Prompt Engineering to Enhance LLM-Based Automated Requirement Traceability and Compliance Checks," 2024, arXiv:2412.08593.

[12] N. Alturayeif, I. Ahmad, and J. Hassine, "TraceLLM: Leveraging large language models with prompt engineering for enhanced requirements traceability," *Requirements Engineering*, vol. 31, no. 1, p. 6, 2026.

[13] G. Long, J. Gong, H. Fang, and T. Chen, "Learning Software Bug Reports: A Systematic Literature Review," *ACM Transactions on Software Engineering and Methodology*, p. 3750040, 2025.

[14] J. P. Meher, S. Biswas, and R. Mall, "Deep learning-based software bug classification," *Information and Software Technology*, vol. 166, p. 107350, 2024.

[15] Q. A. Do, S. R. Chekuri, and T. Bhowmik, "Automated support to capture creative requirements via requirements reuse," in *Proc. 18th Int. Conf. Software Reuse (ICSR)*, 2019, pp. 47–63.

[16] Q. A. Do, T. Bhowmik, and G. L. Bradshaw, "Capturing creative requirements via requirements reuse: A machine learning-based approach," *J. Syst. Softw.*, vol. 170, p. 110730, 2020. [Online]. Available: https://doi.org/10.1016/j.jss.2020.110730

[17] J. T. Almonte, S. A. Boominathan, and N. Nascimento, "Automated non-functional requirements generation in software engineering with large language models: A comparative study," 2025, arXiv:2503.15248.

[18] V. Koscinski, S. Hashemi, and M. Mirakhorli, "On-Demand Security Requirements Synthesis with Relational Generative Adversarial Networks," in *45th IEEE/ACM International Conference on Software Engineering, ICSE 2023, Melbourne, Australia, May 14-20, 2023*. IEEE, 2023, pp. 1609–1621.

[19] F. Wang, C. Arora, Y. Liu, K. Huang, C. Tantithamthavorn, A. Aleti, D. Sambathkumar, and D. Lo, "Multi-Modal Requirements Data-based Acceptance Criteria Generation using LLMs," in *40th IEEE/ACM International Conference on Automated Software Engineering, ASE 2025, Seoul, Korea, Republic of, November 16-20, 2025*. IEEE, 2025, pp. 3334–3345.

[20] M. Krishna, B. Gaur, A. Verma, and P. Jalote, "Using LLMs in Software Requirements Specifications: An Empirical Evaluation," in *2024 IEEE 32nd International Requirements Engineering Conference (RE)*, 2024, pp. 475–483, arXiv:2404.17842.

[21] Z. Nan, Z. Guo, K. Liu, and X. Xia, "Test Intention Guided LLM-Based Unit Test Generation," in *47th IEEE/ACM International Conference on Software Engineering, ICSE 2025*. IEEE, 2025, pp. 1026–1038.

[22] A. Celik and Q. H. Mahmoud, "A Review of Large Language Models for Automated Test Case Generation," *Machine Learning and Knowledge Extraction*, vol. 7, no. 3, p. 97, 2025.

[23] C. Augusto, A. Bertolino, G. De Angelis, F. Lonetti, and J. Morán, "Large Language Models for Software Testing: A Research Roadmap," 2025, arXiv:2509.25043.

[24] D. Jin, Z. Jin, X.-h. Chen, and C. Wang, "MARE: Multi-Agents Collaboration Framework for Requirements Engineering," *CoRR*, 2024, arXiv:2405.03256.

[25] Y. Huang, M. Hu, X. Chen, Z. Jin, and S. Xiao, "Modeling like peeling an onion: Layerwise analysis-driven automatic behavioral model generation," in *Proc. IEEE/ACM 48th Int. Conf. Software Engineering (ICSE)*, 2026.

[26] S. Kirchner and A. C. Knoll, "Generating automotive code: Large language models for software development and verification in safety-critical systems," 2025, arXiv:2506.04038.

[27] A. Bodicoat, G. Jahangirova, and V. Terragni, "Understanding LLM-Driven Test Oracle Generation," 2026, arXiv:2601.05542.

[28] D. Molinelli, L. Di Grazia, A. Martin-Lopez, M. D. Ernst, and M. Pezzè, "Do LLMs Generate Useful Test Oracles? An Empirical Study with an Unbiased Dataset," in *40th IEEE/ACM International Conference on Automated Software Engineering, ASE 2025, Seoul, Korea, November 16–20, 2025*. IEEE, 2025, pp. 278–290.

[29] N. Petrovic, V. Zolfaghari, F. Pan, and A. Knoll, "LLM-empowered functional safety and security by design in automotive systems," 2026, arXiv:2601.02215.

[30] B. V. Balu, F. Geissler, F. Carella, J.-V. Zacchi, J. Jiru, N. Mata, and R. Stolle, "Towards automated safety requirements derivation using agent-based RAG," 2025, arXiv:2504.11243.

[31] "NHTSA office of defects investigation (ODI) complaints and investigations database," National Highway Traffic Safety Administration (NHTSA), Tech. Rep., 2024, available at: https://api.nhtsa.gov/.

[32] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.

[33] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. 2019 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2019, arXiv:1908.10084.

[34] S. Zhao, Y. Shao, Y. Huang, J. Song, Z. Wang, C. Wan, and L. Ma, "Understanding the Fundamental Design Decisions of Retrieval-Augmented Generation Systems," *ACM Transactions on Software Engineering and Methodology*, p. 3802824, 2026, arXiv:2411.19463.

[35] C. Gao, G. Fan, C. Y. Chong, S. Chen, C. Liu, D. Lo, Z. Zheng, and Q. Liao, "A Systematic Literature Review of Code Hallucinations in LLMs: Characterization, Mitigation Methods, Challenges, and Future Directions for Reliable AI," 2025, arXiv:2511.00776.

[36] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[37] R. A. Fisher, "On the interpretation of χ^2 from contingency tables, and the calculation of P," *Journal of the Royal Statistical Society*, vol. 85, no. 1, pp. 87–94, 1922.

[38] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[39] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.

[40] K. L. Gwet, "Computing inter-rater reliability and its variance in the presence of high agreement," *British Journal of Mathematical and Statistical Psychology*, vol. 61, no. 1, pp. 29–48, 2008.

[41] ——, *Handbook of Inter-Rater Reliability: The Definitive Guide to Measuring the Extent of Agreement Among Raters*, 4th ed. Advanced Analytics, LLC, 2014.

[42] A. J. Feinstein and D. V. Cicchetti, "High agreement but low kappa: I. the problems of two paradoxes," *Journal of Clinical Epidemiology*, vol. 43, no. 6, pp. 543–549, 1990.